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ORAL ABSTRACTS

701.EXPERIMENTAL TRANSPLANTATION: BASIC AND TRANSLATIONAL

Increased Fiber Intake Results in Better Overall Survival and Lower GI-aGVHD in Allo-HCT Recipients and Pre-Clinical Gvhd Models

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Background: Dietary fiber can modulate the intestinal microbiome and increase the production of short-chain fatty acids (SCFAs), and bile acids (BAs) which are beneficial microbial metabolites associated with intestinal homeostasis. To investigate the effects of dietary fiber on acute graft-versus-host disease (aGVHD) after allogeneic hematopoietic cell transplantation (allo-HCT), we evaluated a preclinical mouse model of GVHD with defined diet fiber concentrations and analyzed the dietary patterns of allo-HCT patients. We hypothesized that a high-fiber diet in allo-HCT recipients could increase intestinal microbiome diversity, the concentration of beneficial microbial metabolites, and decreased risk of GVHD.

Methods: We collected high-resolution dietary data at Memorial Sloan Kettering Cancer Center (173 allo-HCT patients), from 10 days pre-transplantation to 30 days post-transplantation (3,837 patient days), 16S rRNA sequencing data of fecal samples (1,028 patient days), and gas chromatography-mass spectrometry (GC-MS) data of fecal SCFA concentration for a sub-cohort of acute lower-GI-GVHD patients (110 patient days) and no-GVHD patients (128 patient days) matched by stool sample availability.

We investigated the associations between longitudinal fiber intake and microbial markers (longitudinal data analysis, generalized estimating equations), including α -diversity, butyrate producers, and concentrations of fecal SCFAs between lower-GI-GVHD and no-GVHD patients. To account for high variation in patients' dietary intake trajectories and macronutrients interactions, we performed latent trajectory class analysis to identify patient's dietary intake patterns and to correlate fiber intake and patient survival.

We evaluated how dietary fiber can affect GVHD in a preclinical mouse GVHD model (C57BL/6J into BALB/c) treated with cellulose at 0%, 6%, 12% and 40% concentrations and assessed survival (n=180, n=30 per group), fecal microbial composition by shotgun sequencing (n=165), SCFA concentrations by GC-MS (n=32, n=8 group), immune composition by flow cytometry (n=45, n=15 group) and, the gene expression of cells from the lamina propria of mouse colons by single cell sequencing (scSeq) (n=9, n=3 group).

Results: In allo-HCT patients, a higher fiber intake was significantly associated with a higher microbial α -diversity (p=0.009), a higher abundance of butyrate producers (p=0.03), and a higher concentration of butyrate (p=0.02). Lower GI-GVHD patients had a significantly lower fecal concentrations of butyrate (p=0.03) and acetate (p=0.02) after adjustment for fiber intake. The latent trajectory class analysis identified two patient groups: high fiber (HF) and low fiber (LF) intake, where the HF group had a significantly better overall survival (log-rank p=0.04) and lower cumulative incidence of acute lower GI-GVHD (Gray's p=0.04).

In our preclinical GVHD model, we determined that mice receiving a fiber-rich diet (12% cellulose) had a significant reduction in GVHD lethality (p=0.02), higher microbial α -diversity (p=0.02), decreased relative abundance of the pathogenic bacteria *Enterococcus faecalis* (p=0.001 vs. 0%), a higher ratio of regulatory T cells (CD4+CD25+FOXP3+) over conventional T cells

(CD4+) ($p=0.008$; $p=0.001$) and a higher concentration of cecal butyrate ($p=0.03$ vs. 0%; $p=0.03$ vs. 6%). Importantly, scSeq analysis of CD4+ T cells in the 12% cellulose diet evinced a significant increase in the expression of genes associated with reduced GVHD, including *IDO1* (indoleamine 2, 3-dioxygenase), and *CEACAM1* (carcinoembryonic antigen related cell adhesion molecule 1), as well as the enrichment of the bile acid pathway and associated genes ($p=0.01$ vs. 0%; $p=0.006$ vs. 6%) including the farnesoid X-activated receptor *NR4A1*.

Conclusions: We demonstrated that higher fiber intake in patients after allo-HCT is associated with a) an increase in overall survival; b) a decrease in lower GI-aGVHD cumulative incidence; c) an increase in microbial α -diversity and butyrate-producers; and d) an increase in the fecal concentrations of SCFAs. In addition, the consumption of a 12% cellulose diet in a pre-clinical GVHD model leads to an increase in α -diversity, a decrease in pathogen relative abundance, an increase in Treg/Tconventional ratio, and a decrease in GVHD lethality. These results suggest that dietary fiber could be used in the prevention of GVHD.

Disclosures Peled: *Postbiotics Plus Research*: Current equity holder in publicly-traded company, Membership on an entity's Board of Directors or advisory committees; *Canaccord Genuity, Inc*: Consultancy; *MaaT Pharma*: Consultancy; *Crestone Inc*: Consultancy; *DaVolterra*: Consultancy; *CSL Behring*: Consultancy; *Prodigy Biosciences*: Current equity holder in publicly-traded company, Membership on an entity's Board of Directors or advisory committees; *Seres Therapeutics*: Patents & Royalties, Research Funding.

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